Microsoft 365

MD-100 Windows Client

Module 10 : Troubleshoot the Windows client operating system and apps

Module Agenda

- Employ file recovery in Windows client
- Explore application troubleshooting
- Troubleshoot Windows startup
- Troubleshoot operating system service issues

Lesson 1: Employ file recovery in Windows client

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- Explain the importance of file recovery
- Explain file history
- Protect files using Azure Backup
- Restore files using Backup and Restore (Windows 7)
- Restore files using the Previous Versions tab
- Compare file recovery options
- Troubleshoot file recovery options

Explain the importance of file recovery

Some of the reasons for performing backups

- Protect against accidental file deletion
- Provide recovery from virus infection
- Provide previous versions of files and folders
- Protect against total computer data loss
- Help ensure data availability

Windows features

- Folder Redirection, Offline Files
- Backup and Restore (Windows 7)
- Synchronization with OneDrive
- File History
- Work Folders
- System Image
- Wbadmin.exe
- Copying files

Azure Backup can be used with Windows

Explain file history

File History saves backup copies of user files

Configure and manage by using Control Panel or the Backup section in the Settings app

By default, profile folders and libraries are protected

You can protect additional folders by:

- Adding them to protected libraries
- Using the Backup option in the Settings app

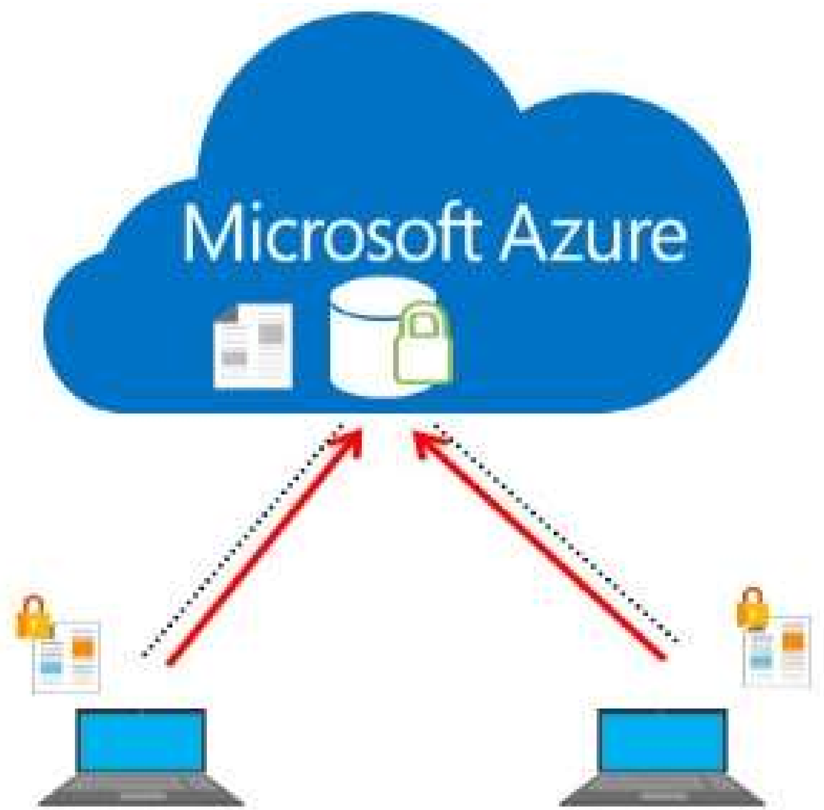
You can save backup copies on a local drive, removable drive, or network location

File History copies protected files hourly and stores copies indefinitely by default

You can preview and restore backup copies:

- You can restore to original or alternate location

Protect files using Azure Backup



Restore files using Backup and Restore (Windows 7)

Graphical backup tool in Windows

Uses Volume Shadow Copy Service for creating backups:

- On local disk, external disk, or network location
- First backup contains all data, later backups contain changes only

Can back up folders, libraries, and volumes

Backup is in .vhdx format

You can use it for creating system image and system repair disk

Creates restore points, used by Previous Versions

Restores data on original or alternate location

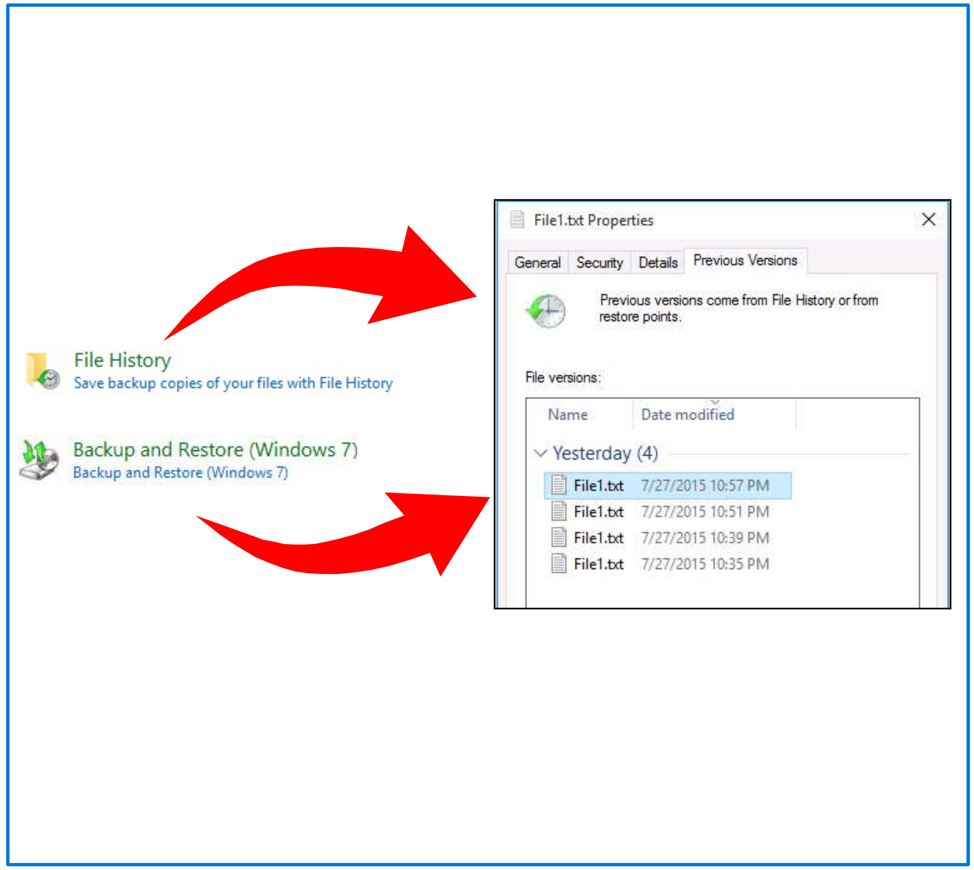
Restore files using the Previous Versions tab

Enables user to view and restore previous versions:

- Files, folders, and volumes

Data comes from File History and restore points:

- Each time when File History runs
- When file is backed up by Backup and Restore (Windows 7)



Compare file recovery options

How often

Types of files

System state

Recovery options

Troubleshoot file recovery options

- Is the tool properly configured?
- Connectivity dependencies
- Target restore device configuration

Knowledge Check

Lesson 2: Explore application troubleshooting

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- Troubleshoot Windows Installer issues
- Resolve desktop app deployment issues
- Identify issues related to desktop app operations
- Resolve issues related to desktop app operations
- Resolve issues related to Universal Windows Platform apps
- Troubleshoot common Microsoft 365 App issues
- Explore the Application Compatibility Toolkit

Troubleshoot Windows Installer issues

1. Verify
2. Msiexec can execute
3. Installer starts without errors
4. Latest updates
5. Re-register Msiexec
6. Reboot
7. Software conflicts

Resolve desktop app deployment issues

- Run as Administrator
- Ensure necessary dependencies are installed
- Leverage the ACT
- Verify AppLocker configuration

Identify issues related to desktop app operations

- Missing OS or application features
- Incorrect app/device/security configuration
- Application errors or performance issues
- Connectivity to dependencies (i.e. database)
- AppLocker blocking

Resolve issues related to desktop app operations

- Install required features
- Repair/Reinstall the applications
- Install application updates/upgrade
- Use tools to identify performance issues
- Reconfigure AppLocker

Resolve issues related to Universal Windows Platform apps

1. UAC settings
2. Security settings
3. Group Policy settings
4. Latest version
5. Clear Microsoft Store cache
6. Application licensing
7. Reinstall

Troubleshoot common Microsoft 365 App issues

Older technologies (i.e. ActiveX, Silverlight, etc.) require Internet Explorer

Favorites are not automatically imported from Internet Explorer

Less commonly used features in Internet Explorer are not always available in Microsoft Edge

Explore the Application Compatibility Toolkit

Use ACT to test and verify your applications:

1. Build a test workstation running Windows 10 and that has all the required apps installed
2. Run the apps to see if there are any issues in functionality or behavior
3. Install ACT on the workstation
4. Open the Compatibility Administrator and run any problematic apps within it
5. Create a custom database to hold test information
6. Create an application fix, if required
7. Save the fix to a distributable location or media
8. Distribute the application fix around your organization

Knowledge Check

Lesson 3: Troubleshoot Windows startup

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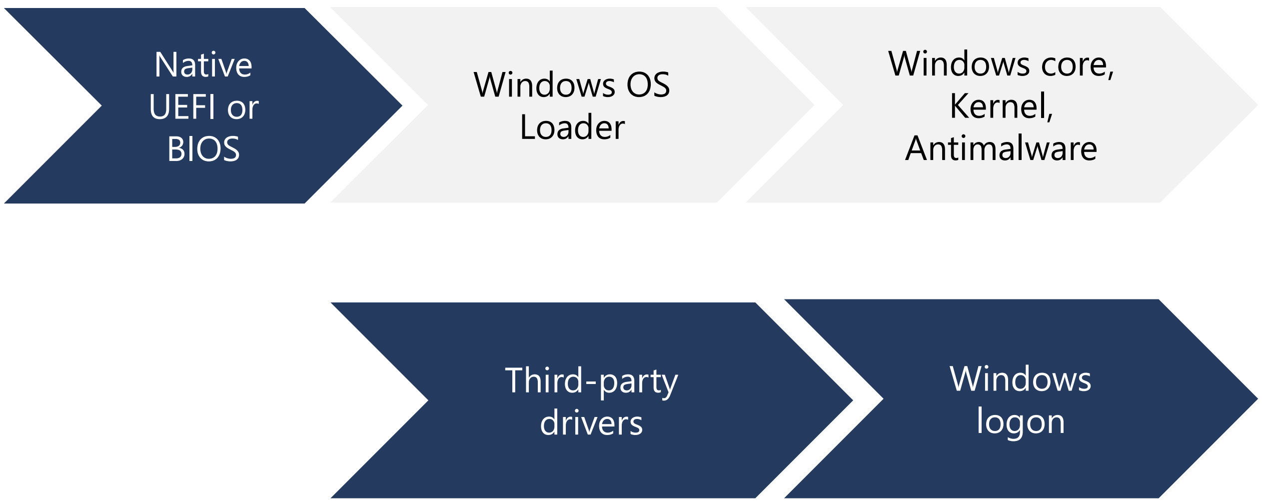
- Explore the Windows client startup architecture
- Compare device recovery procedures
- Use the Windows Recovery Environment
- Explore advanced startup options
- Use the Reset This PC feature
- Examine the System Restore feature
- Explore additional recovery tools in Windows Recovery Environment
- Explore the Windows client Boot Configuration Data store
- Configure the BCD settings

Explore the Windows client startup architecture



Compare device recovery procedures

Operating system is separate from the data:

- You can recover, reinstall or upgrade it without affecting data

Device-recovery features in Windows:

- Driver Roll Back
- System Protection and System Restore
- Startup Recovery
- Reset this PC
- System Image Recovery
- Command Prompt

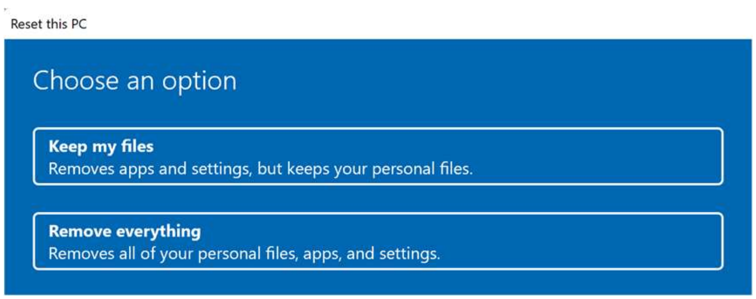
Use the Windows Recovery Environment

Tool	Function
Reset this PC	Lets you choose to keep or remove your files and reinstalls Windows
System Restore	Returns your computer to an earlier state
System Image Recovery	A system image created earlier replaces everything on a computer
Startup Repair	Detects and repairs most common startup issues
Command Prompt	Resolves problems with a service or device driver, and runs diagnostic tools
Go back to the previous build	Preserves personal files, but changes to apps and settings are lost

Explore advanced startup options

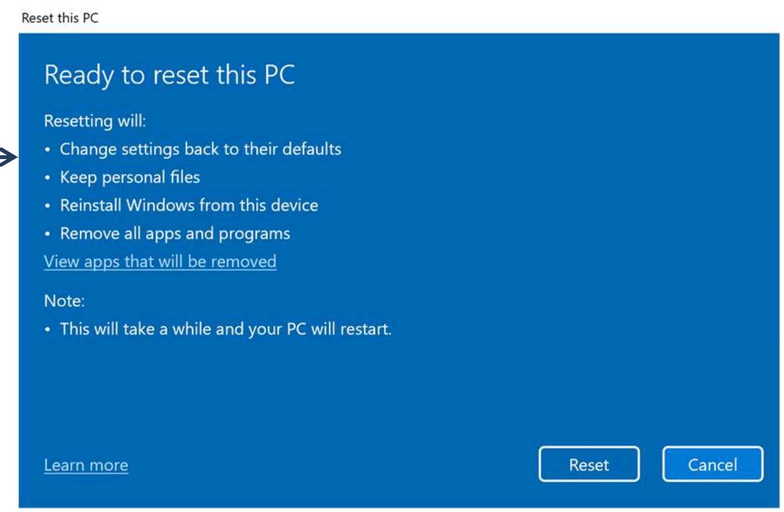
- Enable debugging
- Enable boot logging
- Enable low-resolution video
- Enable Safe Mode
- Enable Safe Mode with Networking
- Enable Safe Mode with Command Prompt
- Disable driver signature enforcement
- Disable early launch anti-malware protection
- Disable automatic restart after failure
- Launch recovery environment

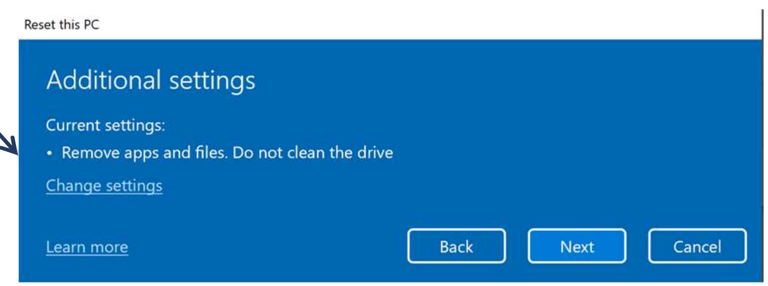
Use the Reset This PC feature



Additional options:

- Keep pre-installed apps
- Clean drive
- Cloud download





Examine the System Restore feature

Create snapshots of computer configuration:

Snapshots are called restore points

Restore points do not include user data

You can use restore points to:

Perform driver rollbacks

Protect against accidental program deletion

Restore Windows configurations to earlier states

Restore points are created before system changes:

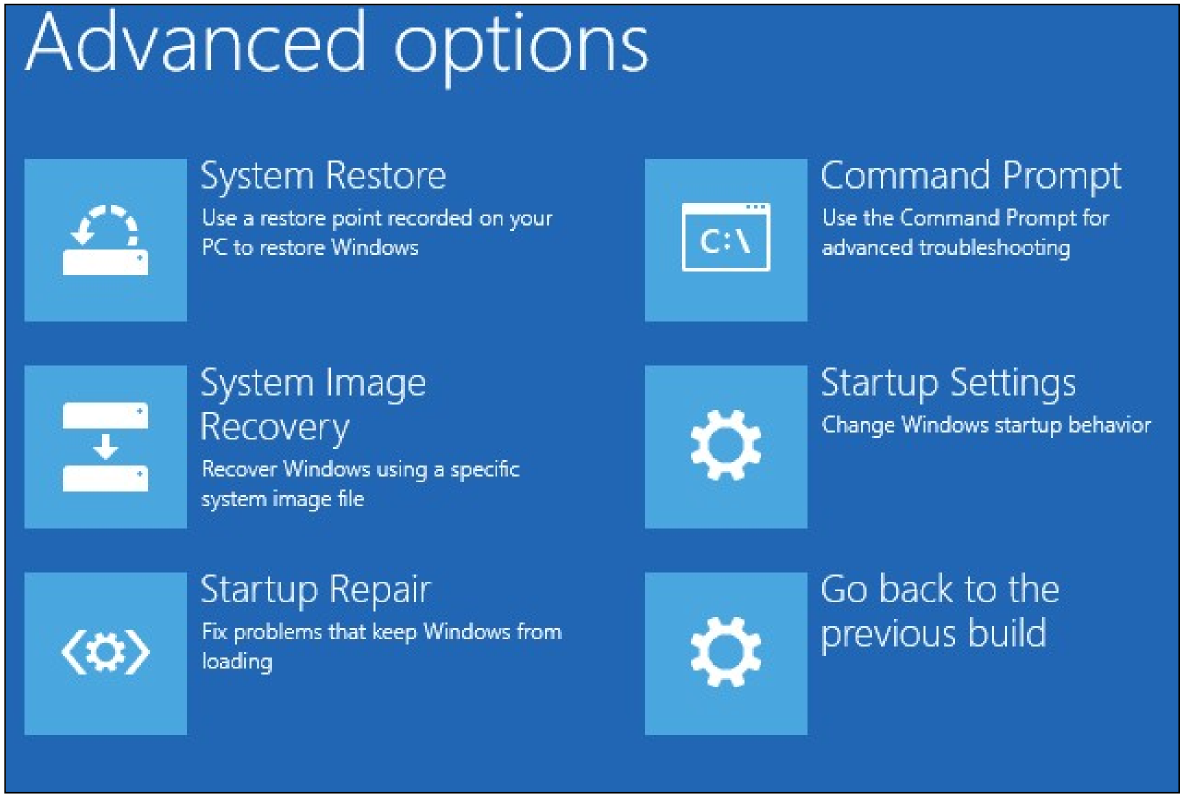
Automatic: app, driver, or Windows updates

Schedule: can be created based on a schedule

Manual: on-demand, before applying restore point

If turned off, all restore points are deleted

Explore additional recovery tools in Windows Recovery Environment



Explore the Windows client Boot Configuration Data store

Database that stores information and configuration about the devices boots to an OS.

Reasons to edit the BCD

- Adding a new hard disk to your Windows computer and changing the logical drive numbering.
- Installing additional operating systems on your Windows computer to create a multiboot configuration.
- Deploying Windows to a new computer with a blank hard disk, requiring you to configure the appropriate boot store.
- Performing a backup or restoring a corrupted BCD.

Configure the BCD settings

Modify with:

- Startup and Recovery
- MSConfig
- BCDEdit
- BootRec

Knowledge Check

Lesson 4: Troubleshoot operating system service issues

Lesson 4: Troubleshoot operating system service issues

- Examine operating system services
- Identify failed operating system services
- Disable operating system services
- Troubleshoot locked accounts in Windows clients
- Identify sign-in errors

Examine operating system services

Load and run in the background without user intervention

Support application requests, for example, when an application needs to open a file, it relies on a system service to retrieve that file from the disk

Can make calls to device drivers when a request is sent to a physical device

Identify failed operating system services

Windows provides a number of ways of locating service-related problems:

- Event Viewer
- Log files
- Stop codes
- Action Center

Disable operating system services

Depending on the circumstances, you can disable a service in one of the following ways:

- Safe mode
- Command Prompt
- System Configuration tool

Troubleshoot locked accounts in Windows clients

Common causes for repeated lockouts:

- Cached credentials used by application
- Service account passwords
- Persistent drive mappings

Identify sign-in errors

Verify sign in error or message

Unlock account in Active Directory

Review event logs

Knowledge Check

Practice Labs

Using File History to Recover Files

Using Windows RE and Advanced Startup to recover from Boot Failures

Troubleshooting Boot Failure with the BCD store

Recovering Windows using Reset This PC

Module Review

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